Lab 5: Static Code Analysis

School Bus Transport Register - commands and terminal output

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Commands used

```
python school_transport.py
pip install pylint bandit flake8
pylint school_transport.py > pylint_report.txt
bandit -r school_transport.py > bandit_report.txt
flake8 school_transport.py > flake8_report.txt
pylint school_transport.py > pylint_after.txt
bandit -r school_transport.py > bandit_after.txt
flake8 school_transport.py > flake8_after.txt
```

Part 1 - Original code (baseline)

1. Run the program

```
PS C:\Users\Aryan\Desktop\L5_24_school_transport> python school_transport.py
dispatching bus R1
['Zoya']
```

2. Install the analysis tools

```
PS C:\Users\Aryan\Desktop\L5_24_school_transport> pip install pylint bandit flake8
Requirement already satisfied: pylint in
c:\users\aryan\appdata\local\programs\python\python313\lib\site-packages (4.1.3)
Requirement already satisfied: bandit in
c:\users\aryan\appdata\local\programs\python\python313\lib\site-packages (1.9.4)
Requirement already satisfied: flake8 in
c:\users\aryan\appdata\local\programs\python\python313\lib\site-packages (7.4.1)
Requirement already satisfied: astroid<=4.4,>=4.3.4 in
c:\users\aryan\appdata\local\programs\python\python313\lib\site-packages (from pylint) (4.3.4)
Requirement already satisfied: colorama>=0.4.5 in
c:\users\aryan\appdata\roaming\python\python313\site-packages (from pylint) (0.4.6)
Requirement already satisfied: dill>=0.3.6 in
c:\users\aryan\appdata\local\programs\python\python313\lib\site-packages (from pylint) (0.4.1)
Requirement already satisfied: isort!=5.13,<10,>=5 in
c:\users\aryan\appdata\local\programs\python\python313\lib\site-packages (from pylint) (9.0.2)
Requirement already satisfied: mccabe<0.8,>=0.6 in
c:\users\aryan\appdata\local\programs\python\python313\lib\site-packages (from pylint) (0.7.0)
Requirement already satisfied: platformdirs>=2.2 in
c:\users\aryan\appdata\roaming\python\python313\site-packages (from pylint) (4.3.8)
Requirement already satisfied: tomlkit>=0.10.1 in
c:\users\aryan\appdata\local\programs\python\python313\lib\site-packages (from pylint) (0.15.1)
Requirement already satisfied: mypy-extensions>=1.1.0 in
c:\users\aryan\appdata\local\programs\python\python313\lib\site-packages (from
isort!=5.13,<10,>=5->pylint) (1.1.0)
Requirement already satisfied: PyYAML>=5.3.1 in
c:\users\aryan\appdata\local\programs\python\python313\lib\site-packages (from bandit) (6.0.3)
Requirement already satisfied: stevedore>=1.20.0 in
c:\users\aryan\appdata\local\programs\python\python313\lib\site-packages (from bandit) (5.9.1)
Requirement already satisfied: rich in
c:\users\aryan\appdata\local\programs\python\python313\lib\site-packages (from bandit) (15.0.0)
Requirement already satisfied: pycodestyle<2.16.0,>=2.15.0 in
c:\users\aryan\appdata\local\programs\python\python313\lib\site-packages (from flake8) (2.15.0)
Requirement already satisfied: pyflakes<4.1.0,>=4.0.0 in
c:\users\aryan\appdata\local\programs\python\python313\lib\site-packages (from flake8) (4.0.3)
Requirement already satisfied: markdown-it-py>=2.2.0 in
c:\users\aryan\appdata\local\programs\python\python313\lib\site-packages (from rich->bandit) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
c:\users\aryan\appdata\roaming\python\python313\site-packages (from rich->bandit) (2.19.2)
```

```
Requirement already satisfied: mdurl~=0.1 in
c:\users\aryan\appdata\local\programs\python\python313\lib\site-packages (from markdown-it-
py>=2.2.0->rich->bandit) (0.1.2)
```

3. Pylint baseline

```
PS C:\Users\Aryan\Desktop\L5_24_school_transport> pylint school_transport.py > pylint_report.txt
PS C:\Users\Aryan\Desktop\L5_24_school_transport> Get-Content pylint_report.txt
************* Module school_transport
school_transport.py:5:0: C0301: Line too long (110/100) (line-too-long)
school_transport.py:9:0: C0301: Line too long (116/100) (line-too-long)
school_transport.py:1:0: C0114: Missing module docstring (missing-module-docstring)
school_transport.py:8:0: C0116: Missing function or method docstring (missing-function-docstring)
school_transport.py:8:0: C0103: Function name "enrollRider" doesn't conform to snake_case naming
style (invalid-name)
school_transport.py:13:0: C0116: Missing function or method docstring (missing-function-docstring)
school_transport.py:13:0: C0103: Function name "routeRoster" doesn't conform to snake_case naming
style (invalid-name)
school_transport.py:13:23: W0613: Unused argument 'driver' (unused-argument)
school_transport.py:17:0: C0116: Missing function or method docstring (missing-function-docstring)
school_transport.py:17:0: C0103: Function name "exportRoster" doesn't conform to snake_case naming
style (invalid-name)
school_transport.py:18:10: W1514: Using open without explicitly specifying an encoding (unspecified-
encoding)
school_transport.py:18:10: R1732: Consider using 'with' for resource-allocating operations
(consider-using-with)
school_transport.py:23:0: C0116: Missing function or method docstring (missing-function-docstring)
school_transport.py:23:0: C0103: Function name "dispatchAlert" doesn't conform to snake_case naming
style (invalid-name)
school_transport.py:27:0: C0116: Missing function or method docstring (missing-function-docstring)
school_transport.py:27:0: C0103: Function name "feeCollected" doesn't conform to snake_case naming
style (invalid-name)
school_transport.py:30:4: W0702: No exception type(s) specified (bare-except)
school_transport.py:34:0: C0116: Missing function or method docstring (missing-function-docstring)

-----------------------------------
Your code has been rated at 4.00/10
```

4. Bandit baseline

```
PS C:\Users\Aryan\Desktop\L5_24_school_transport> bandit -r school_transport.py > bandit_report.txt
PS C:\Users\Aryan\Desktop\L5_24_school_transport> Get-Content bandit_report.txt
Run started:2026-10-14 06:49:31.213423+00:00

Test results:
>> Issue: [B404:blacklist] Consider possible security implications associated with the subprocess
module.
   Severity: Low   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_imports.html#b404-import-
subprocess
   Location: .\school_transport.py:1:0
1   import subprocess
2   import sqlite3
3

--------------------------------------------------
>> Issue: [B608:hardcoded_sql_expressions] Possible SQL injection vector through string-based query
construction.
   Severity: Medium   Confidence: Medium
   CWE: CWE-89 (https://cwe.mitre.org/data/definitions/89.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b608_hardcoded_sql_expressions.html
   Location: .\school_transport.py:9:19
8   def enrollRider(student,route,stop,fee):
```

```
9       conn.execute(f"INSERT INTO riders (student, route, stop, fee) VALUES ('{student}',
'{route}', '{stop}', {fee})")
10      conn.commit()

--------------------------------------------------
>> Issue: [B602:subprocess_popen_with_shell_equals_true] subprocess call with shell=True identified,
 security issue.
   Severity: High   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info:
https://bandit.readthedocs.io/en/1.9.4/plugins/b602_subprocess_popen_with_shell_equals_true.html
   Location: .\school_transport.py:24:4
23  def dispatchAlert(route):
24      subprocess.run("echo dispatching bus " + route, shell=True, check=False)
25

--------------------------------------------------
>> Issue: [B101:assert_used] Use of assert detected. The enclosed code will be removed when
compiling to optimised byte code.
   Severity: Low   Confidence: High
   CWE: CWE-703 (https://cwe.mitre.org/data/definitions/703.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b101_assert_used.html
   Location: .\school_transport.py:38:4
37      enrollRider("Zoya", "R2", "Fort Gate", 2100)
38      assert feeCollected("R1") == 3600
39      exportRoster("route_r1.txt", "R1")

--------------------------------------------------

Code scanned:
    Total lines of code: 30
    Total lines skipped (#nosec): 0
    Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
    Total issues (by severity):
        Undefined: 0
        Low: 2
        Medium: 1
        High: 1
    Total issues (by confidence):
        Undefined: 0
        Low: 0
        Medium: 1
        High: 3
Files skipped (0):
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.13.5
```

5. Flake8 baseline

```
PS C:\Users\Aryan\Desktop\L5_24_school_transport> flake8 school_transport.py > flake8_report.txt
PS C:\Users\Aryan\Desktop\L5_24_school_transport> Get-Content flake8_report.txt
school_transport.py:5:80: E501 line too long (110 > 79 characters)
school_transport.py:8:24: E231 missing whitespace after ','
school_transport.py:8:30: E231 missing whitespace after ','
school_transport.py:8:35: E231 missing whitespace after ','
school_transport.py:9:80: E501 line too long (116 > 79 characters)
school_transport.py:14:80: E501 line too long (95 > 79 characters)
school_transport.py:23:1: E302 expected 2 blank lines, found 1
school_transport.py:29:80: E501 line too long (98 > 79 characters)
school_transport.py:30:5: E722 do not use bare 'except'
```

Part 2 - Fixed code (after changes)

6. Run the fixed program

```
PS C:\Users\Aryan\Desktop\L5_24_school_transport> python school_transport.py
dispatching bus R1
['Zoya']
PS C:\Users\Aryan\Desktop\L5_24_school_transport> Get-Content route_r1.txt
Tara
Kabir
```

7. Pylint after fixes

```
PS C:\Users\Aryan\Desktop\L5_24_school_transport> pylint school_transport.py > pylint_after.txt
PS C:\Users\Aryan\Desktop\L5_24_school_transport> Get-Content pylint_after.txt

-----------------------------------
Your code has been rated at 10.00/10
```

8. Bandit after fixes

```
PS C:\Users\Aryan\Desktop\L5_24_school_transport> bandit -r school_transport.py > bandit_after.txt
PS C:\Users\Aryan\Desktop\L5_24_school_transport> Get-Content bandit_after.txt
Run started:2026-10-14 06:49:33.070843+00:00

Test results:
    No issues identified.

Code scanned:
    Total lines of code: 50
    Total lines skipped (#nosec): 0
    Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
    Total issues (by severity):
        Undefined: 0
        Low: 0
        Medium: 0
        High: 0
    Total issues (by confidence):
        Undefined: 0
        Low: 0
        Medium: 0
        High: 0
Files skipped (0):
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.13.5
```

9. Flake8 after fixes

```
PS C:\Users\Aryan\Desktop\L5_24_school_transport> flake8 school_transport.py > flake8_after.txt
PS C:\Users\Aryan\Desktop\L5_24_school_transport> Get-Content flake8_after.txt
```